Mortgage Prepayment Risk Modeling

From Kaplan–Meier to Deep Survival Analysis

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Project Scope

Goal. Build a survival-analysis pipeline for mortgage prepayment on Freddie Mac single-family data, comparing classical, ML, and deep-learning approaches.

What we implemented:

- **Part A.** Kaplan–Meier survival on the full 31.7M-loan population, stratified by FICO/LTV/vintage/purpose
- **Part B.** Cox proportional hazards (origination + vintage-macro covariates), PH diagnostics
- **Part C.** ML models (LogReg, RF, LightGBM) for horizon-specific prepayment prediction; Cox baseline
- **Part D.** Deep Cox model (DeepSurv) + linear-Cox-via-SGD ablation

Headline result. Deep Cox: AUC = 0.729 at T=60 vs 0.685 for the matched-architecture linear baseline. Gap widens monotonically with horizon: +0.003 at T=12 → +0.045 at T=60.

Data: Freddie Mac Single-Family Loan-Level

Population. 31,688,129 30-year fixed-rate mortgages, originated 1999–2025.

Outcomes (cause-specific framing for prepayment):

- Prepaid (Zero Balance Code 01): 21,748,774 (68.6%) — the event
- Credit-related termination (ZBC 03, 09): 532,230 (1.7%)
- Other termination (ZBC 02, 15, 16, 96): 265,311 (0.8%)
- Right-censored (still active): 9,141,814 (28.8%)

Survival table. One row per loan with `Duration`, `Event_Prepay`, and 39 origination covariates. Built once via polars, persisted to parquet. Sentinel codes (DTI=999, MI=999, Credit=9999) cleaned to NaN.

Train/test split. By vintage — train: $\leq$ 2014 (full crisis cycle); test: 2015–2019. Single `canonical_split.parquet` is the source of truth shared by Parts C and D.

Pipeline Architecture

Six scripts + four plotting notebooks, end-to-end $\approx$ 50 minutes on CPU.

Stage	Output	Runtime
`build_survival_table.py`	`survival_table.parquet`	8 min
`part_a_survival_analysis.py`	`results_a/*.parquet`	3.5 min
`part_b_cox_model.py`	`results_b/*.parquet`	12 min
`part_c_ml_models.py`	`results_cd/*.parquet`	12 min
`part_d_deep_cox.py`	`results_cd/D_*.parquet`	19 min

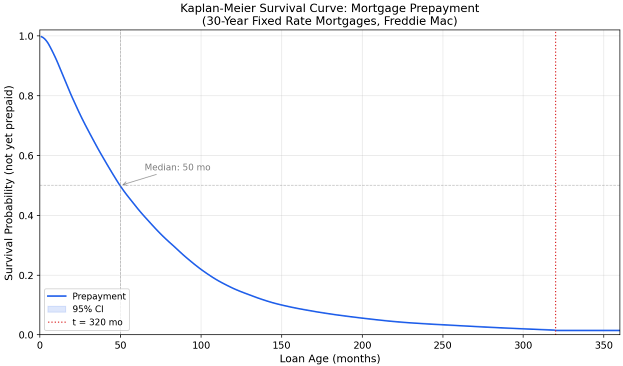
Design principle: compute / plot separation.

- Scripts emit parquet only; no plotting in compute scripts.
- Plotting notebooks live alongside their parquets and read from `..`
- A central `utilities.py` hosts the canonical pre-processing functions (`build_feature_matrix`, `build_horizon_targets`, `standardize_three`) used by all parts.

Reproducibility. Single-source-of-truth split; encoders fit once on train, persisted; deterministic seeds throughout.

Part A — Kaplan–Meier on the Full Population

No sampling: KM fit on all 31.7M loans.



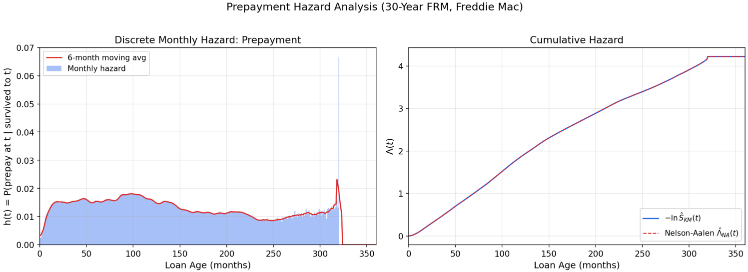
Key timepoints:

t (mo)	$S(t)$	% prepaid
12	0.898	10.2
36	0.623	37.7
60	0.427	57.3
120	0.155	84.5
240	0.037	96.3

Median time to prepayment: **50 months**.

All non-prepay outcomes pooled as right-censored (cause-specific framing): 9.94M censored vs 21.75M events.

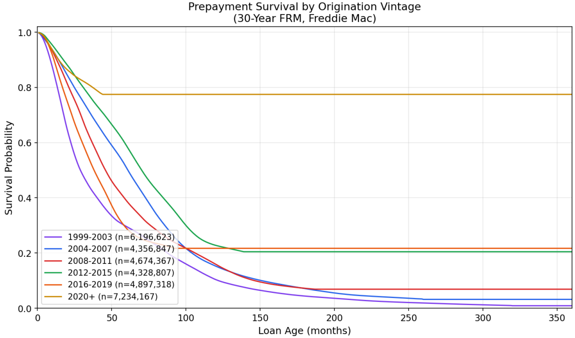
Part A — Hazards

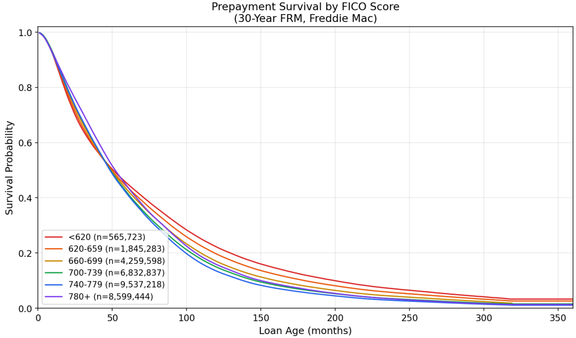


Discrete monthly hazard: ramps from $\approx 1\%$ at $t=6$ to $\approx 1.4\%$ at $t=12$, then plateaus at 1.5–1.7% (annualized $\approx$ 17–18%) from year 2 onward.

Sanity check: $-\log S_{\mathrm{KM}}$ vs Nelson–Aalen cumulative hazard agree to within numerical precision (as expected — both estimate $H(t)$).

Part A — Stratification by Vintage and FICO





Vintage dominates. 1999–2003: 96% prepay, median 29 mo (post-2000 rate cuts). 2020+: only 16% prepay, median undefined (high-rate environment, recent vintages).

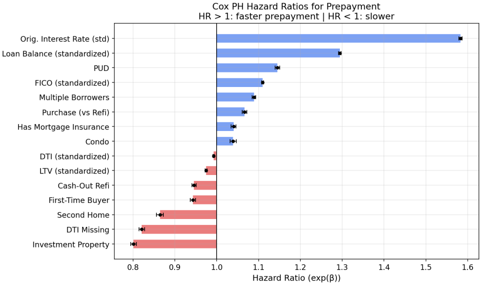
FICO is monotone but modest. Prepay rate ranges from 78% (sub-620) to 65% (780+). Higher-FICO borrowers prepay less in proportion — they often refinance into shorter-term loans, which exit the 30y FRM population sooner.

Part B — Cox PH with Origination Covariates

Base model. 5% sample (1.58M loans, 1.08M events), 15 origination covariates, lifelines + ℓ_2 penalizer 0.001.
Concordance index: 0.6257.

Largest hazard ratios:

- **Original interest rate** (z): HR = 1.58
  the dominant driver — high-rate borrowers refi when rates fall
- **Original UPB** (z): HR = 1.29
- **Investment property**: HR = 0.80
  investors prepay less
- **DTI missing**: HR = 0.82
  Relief Refinance flag (FAQ Q67)
- **FICO** (z): HR = 1.11

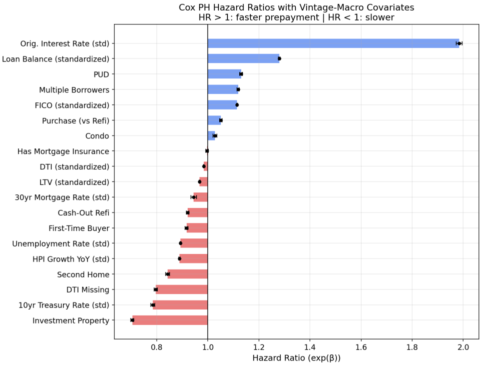


Macro covariates joined at `FirstPaymentDate` (origination): MortgageRate, 10y Treasury, Unemployment, HPI YoY.
Concordance: 0.6428 ($\Delta C = +0.017$, ΔAIC $= -28{,}510$).

Macro hazard ratios:

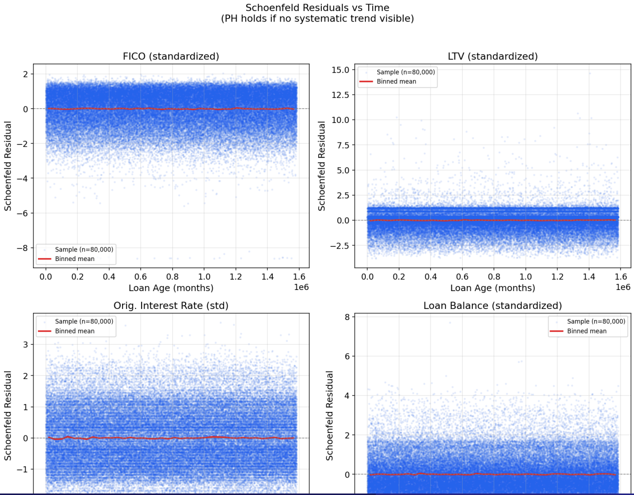
- **10y Treasury at origination**: HR $= 0.78$
- **HPI YoY at origination**: HR $= 0.89$
- **Unemployment at origination**: HR $= 0.89$
- **30y mortgage rate at origination**: HR $= 0.94$

Caveat: these are vintage-macro covariates — the macro state *when the loan was originated*, not time-varying through the loan's life. A truly dynamic Cox would need the loan-month panel.



Part B — Proportional Hazards Diagnostic

Schoenfeld residual test rejects PH for 12 of 15 covariates at $\alpha = 0.05$.



Why this matters.

- Standard finding on mortgage data; this is the principal motivation for moving beyond Cox.
- PH violation means the hazard ratio for, say, FICO is *not constant* over the life of the loan — the relative effect of credit quality on prepayment behaves differently at year 1 vs year 5.
- Linear models can't represent this. Tree-based and deep models can, by interacting features with loan age.

Part C — ML Models for Horizon-Specific Prediction

Reframing. For each horizon $T \in \{12, 24, 36, 60\}$, we model:

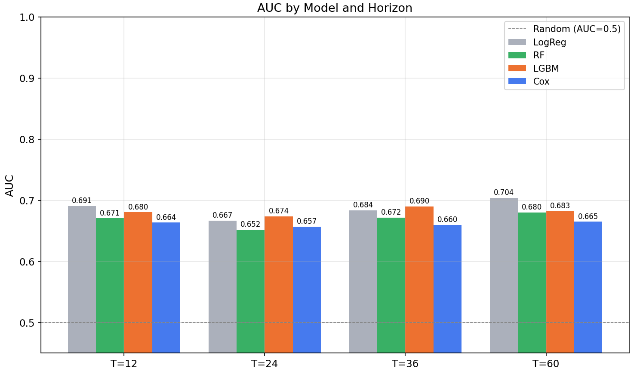
$$P(\text{prepaid by } T \mid X) \quad \text{as a binary classification.}$$

Cause-specific exclusions. Loans with competing termination at Duration $\leq T$ or right-censored before T are excluded (NaN target), not labeled as 0.
Setup: 10% sample, 1.85M train / 597k test, 68 features after one-hot. Same canonical split as Part D.
Four models:

- Logistic Regression (ℓ_2, internal scaler)
- Random Forest (200 trees)
- LightGBM (with held-out-validation early stopping)
- Cox baseline (lifelines, predict $1 - S(T \mid X)$)

Evaluated: AUC, Brier, log-loss, accuracy — per model, per horizon, stratified by vintage and FICO.

Part C — Out-of-Sample AUC

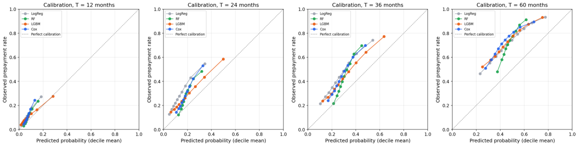


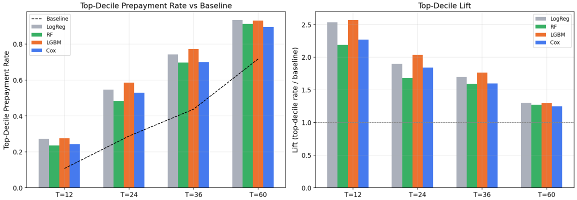
Model	T=12	T=24	T=36	T=60
Cox	0.66	0.66	0.66	0.67
RF	0.67	0.65	0.67	0.68
LGBM	0.68	0.67	0.69	0.68
LogReg	0.69	0.67	0.68	0.70

Observations:

- All four models in the 0.65–0.70 band
- LogReg is competitive, even best at T=12 and T=60
- Marginal effects in the data are largely linear-additive
- Tree models' axis-aligned splits don't unlock much extra signal (yet)

Part C — Calibration and Top-Decile Lift





Calibration is reasonable for all models. LightGBM and LogReg sit closest to the diagonal; Cox is slightly under-confident at the high end.

Top-decile lift. At T=12 the top decile catches 27.6% prepayments (LightGBM) vs 10.75% baseline → **lift = 2.57**×. Saturates at T=60 because the base rate is already 71.7%.

Part D — Deep Cox Architecture and Training

Model: $\lambda(t \mid X) = \lambda_0(t) \cdot \exp\big(f_\theta(X)\big)$ where f_θ is a feed-forward network.

Architecture (DeepSurv-standard):

- 3 hidden layers $\times$ 64 ReLU units
- Batch normalization, dropout 0.1
- No output bias
- 13,184 parameters

Training (pycox + torchtuples):

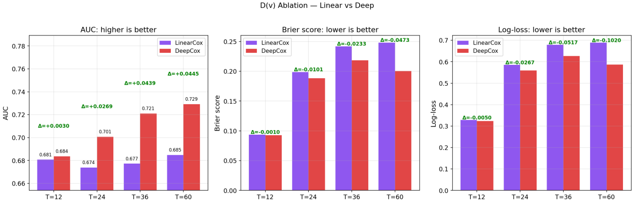
- Adam, $\eta = 10^{-3}$, batch 512
- Up to 50 epochs, ES patience 8 on val partial-likelihood loss
- Mini-batch Cox approximation (risk sets batch-local — standard pycox / DeepSurv)
- Breslow baseline post-fit on full train

Linear baseline (D(v) ablation):

- Same pycox infrastructure
- `num_nodes=[]` $\rightarrow$ no hidden layers
- 68 parameters (one per feature)
- Functionally a SGD-fit classical Cox

Why this is the cleanest possible ablation: same data, same optimizer, same baseline-hazard estimator $\rightarrow$ the only difference is depth.
Runtime: 17 min Deep + 1 min Linear on CPU.

Part D — Linear vs Deep Ablation (D(v))

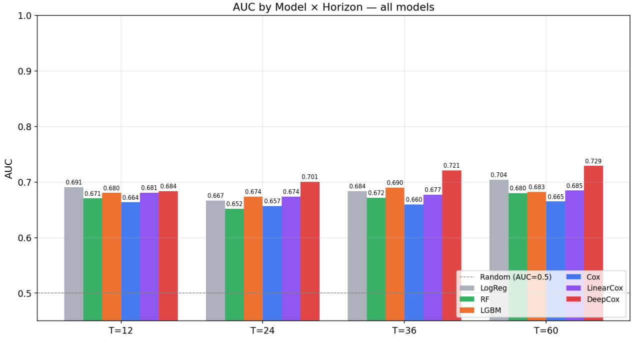


T	Linear	Deep	ΔAUC
12	0.6807	0.6837	+0.003
24	0.6738	0.7007	+0.027
36	0.6772	0.7211	+0.044
60	0.6848	0.7292	+0.045

The gap widens monotonically with horizon.

Sadhwani et al. (2021) predicted exactly this — nonlinear interactions accumulate over the loan's life.

Part D — Head-to-Head with All Models



Model	T=12	T=24	T=36	T=60
Cox	.66	.66	.66	.67
RF	.67	.65	.67	.68
LGBM	.68	.67	.69	.68
LinearCox	.68	.67	.68	.68
LogReg	.69	.67	.68	.70
DeepCox	**.68**	**.70**	**.72**	**.73**

DeepCox is best at every horizon $\geq$ 24 months, often by 0.04–0.05 AUC.
At T=12 LogReg edges it slightly — short-horizon prepay is dominated by "easy" cases (large rate-incentive borrowers).

Why Deep Wins at Long Horizons

Linear assumption: the log-hazard is additive in covariate effects.

$$\log \lambda(t \mid X) \;=\; \log \lambda_0(t) + \beta_1 x_1 + \beta_2 x_2 + \cdots$$

A loan with high rate *plus* high FICO *plus* low LTV experiences a hazard that is the product of independently-acting multipliers.

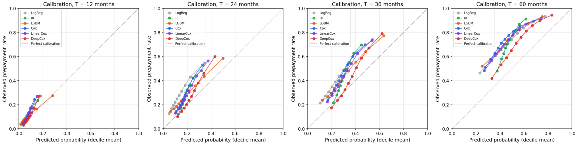
Reality: prepayment incentive is *joint* in covariates.
The refinancing decision conditional on still being alive at T=36 depends on the borrower's current FICO, balance, rate environment they would face refinancing into — and these interact non-trivially.

At long horizon, more interaction paths matter. The deep network captures them; the linear model collapses them. Hence the widening gap.

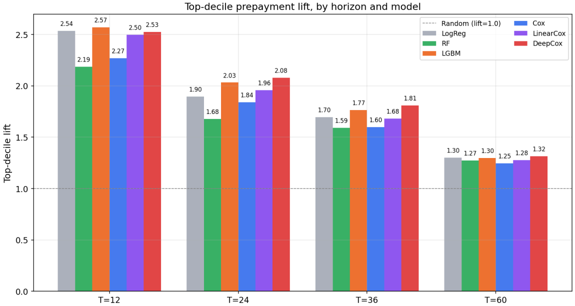
At T=12 only borrowers with very strong, singular incentive have prepaid. Easy cases for any model.

Same conclusion as Sadhwani–Giesecke–Sirignano (2021) on the much larger CoreLogic dataset.

Part D — Calibration and Top-Decile Lift



Calibration. DeepCox sits close to the diagonal at all horizons. LinearCox slightly under-confident.



Top-decile lift. At moderate horizons (T=24, T=36) DeepCox catches more positives:

- T=24: DeepCox 2.08× vs LGBM 2.03× vs LogReg 1.90×
- T=36: DeepCox 1.81× vs LGBM 1.77×

Limitations and Future Work

What this study didn't do:

1. **Time-varying covariates.** Macro covariates are joined at origination only. A truly dynamic Cox would join the current rate environment, current LTV (HPI-updated), current unemployment — but requires the loan-month panel ($\sim$3.5B rows).
2. **Competing risks.** We use cause-specific censoring for prepayment. A multi-output network jointly modeling prepay, default, and survival (Fine–Gray subdistribution hazards or multi-state) would be more complete.
3. **Mini-batch Cox sensitivity.** Risk sets are batch-local; a sensitivity check across batch sizes 256/512/1024/2048 would strengthen the conclusions. Standard practice in DeepSurv literature, but worth verifying.
4. **Distribution shift.** Vintage-based train/test split means the test set sees a different rate environment than train. Honest — but the absolute AUCs reflect distribution shift, not just model capacity.

Natural extensions:

- Time-dependent covariates via loan-month panel
- Recurrent or attention-based survival models
- Interest-rate scenario analysis using the trained model

Key Takeaways

1. **Mortgage prepayment is not a linear-in-covariates process.** Schoenfeld test rejects PH for 12/15 covariates. The cleanest evidence: Linear vs Deep ablation at fixed everything-else.
2. **Deep Cox AUC = 0.729 at T=60** vs 0.685 linear baseline ($\Delta = +0.045$). Not a small effect. And it grows with horizon.
3. **Tree models don't close the gap.** LightGBM tops out at 0.68–0.69; Random Forest below that. Suggests interactions are smooth-and-multivariate (good for MLPs) rather than coarse-and-axis-aligned (good for trees).
4. **Logistic regression is surprisingly competitive at the short end** (T=12). When prepayment is concentrated in a tail of obvious refi candidates, marginal effects suffice.
5. **Vintage-macro covariates earn their place.** Cox concordance jumps from 0.626 to 0.643 just from joining FRED data at origination.
6. **Deep Cox is well-calibrated** — rare for a survival NN, attributable to the structural Breslow baseline rather than direct probability fitting.

Code Statistics and Process Notes

Codebase:

Subsamples and splits:

Part A	Full 31.69M-loan population; no sampling.
Part B	5% Cox sample: 1.58M loans, 1.08M events; PH diagnostic uses 158k rows.
Part C	Shared 10% universe: 1.85M train / 597k test; 68 one-hot features.
Part D	Same C/D split; 10% validation carve: 1.66M inner train / 185k val / 597k test.

Engineering notes:

- **Memory.** Polars survival-table build, sentinel cleaning, and float32 kept peak around 10 GB.
- **Reproducibility.** One `canonical_split.parquet`; train-only encoders/scalers persisted and reused by Parts C/D.
- **Leakage fixes.** Cross-AI review caught encoder fitting before split and Part D validation carving after standardization; both corrected.
- **Session boundaries.** `utilities.py` acted as the API contract across multiple coding sessions.
- **Model plumbing.** Thin pycox-to-lifelines adapter enabled Deep/Linear Cox vs classical Cox comparison.

Thank you

Questions?

Reproducibility

Six scripts + four notebooks, $\approx$ 50 min CPU runtime.
All encoders, scalers, and seeds persisted.

References

Sadhwani, Giesecke, Sirignano (2021), *J. Financial Econometrics*
Katzman et al. (2018), DeepSurv, *BMC Medical Research Methodology*
Kvamme et al. (2019), pycox, *JMLR*
Davidson-Pilon (2019), lifelines, *JOSS*